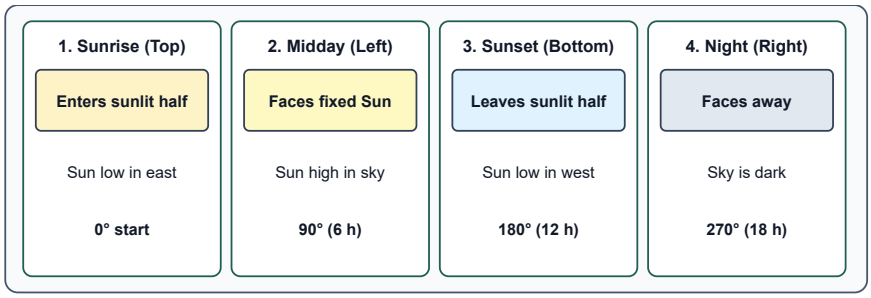
Reference sheets

Reference A · One marked place rotating through light and dark (North-Pole view)

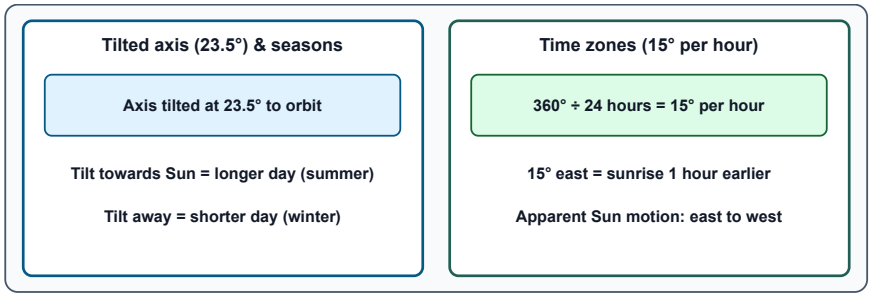


Four positions of a marked place in the North-Pole view with sunlight from the left: 1. Sunrise, 2. Midday, 3. Sunset, 4. Night.

Reference only. Do not write on this sheet. Keep the Sun fixed on the left and follow the red marker anticlockwise (1 → 2 → 3 → 4).

Position in North-Pole view	Time of day	Space viewpoint clue (what the marker does)	Observer viewpoint clue (how the sky looks)
Position 1 · Top boundary	Sunrise	Turns anticlockwise from the unlit half into the sunlit left half	Sun appears low in the eastern sky
Position 2 · Left sunlit side	Middle of the day	Faces directly towards the fixed Sun on the left	Sun appears high across the sky
Position 3 · Bottom boundary	Sunset	Turns anticlockwise out of the sunlit half into the unlit right half	Sun appears low in the western sky
Position 4 · Right unlit side	Night	Faces away from the fixed Sun inside the shaded right half	Sky is dark; the Sun is below the horizon

Reference B · Key Stage 3: tilted axis (23.5°), day length, seasons, and time zones



Key Stage 3 model: Earth's 23.5° tilted axis causes changing day length and seasons, while 360° rotation in 24 hours (15° per hour) causes time zones.

Key Stage 3 concept	Space mechanism (actual cause)	Observable effect on Earth
Apparent Sun motion	Earth rotates west to east on its axis once every 24 hours	The observer turns with Earth, so the Sun appears to move from east to west across the sky
Time zones (15° per hour)	Earth rotates 360° in 24 hours, turning 15° of longitude each hour (360° ÷ 24 = 15°)	Global time zones shift by 1 hour for every 15° of longitude east or west
Tilted axis & length of day	Earth's axis is tilted at 23.5° relative to its orbit around the Sun	Daylight is longer than 12 hours when our hemisphere tilts towards the Sun and shorter when it tilts away
Seasons	Earth orbits the Sun once a year while keeping its 23.5° axial tilt pointed in the same direction	Hemisphere tilted towards the Sun has summer (longer days, higher Sun); opposite hemisphere has winter (shorter days, lower Sun)

Made by Matt

Tier pack · Supported

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Look at the arrival picture cue. Which letter marks the place on the bright sunlit half of Earth?

Help: Look at the yellow sunlight arrows on the left of the drawing and check the letter on the lit half.

2. Look at Marker Q on the shaded right half of the drawing. Why is no sunlight reaching Marker Q?

Help: Follow the sunlight arrows from the left and see which side of Earth blocks the light.

3. Think back: what do we call the imaginary pole-to-pole line that Earth spins around?

Help: It is a four-letter science word starting with a.

4. In a classroom model, what object does the fixed lamp represent?

Help: Think about which object in space gives out light for day.

You do tasks

1. Use Reference A (North-Pole view with the fixed Sun on the left). Order the four daily moments starting at Position 1: Night, Sunset, Sunrise, Middle of the day.

Staff I ☐ S ☐ C ☐

Help: Start at Position 1 (top boundary, entering light) and follow the anticlockwise arrow around Earth.

2. Complete these sentences using the word bank (day, night, rotates): When our place faces the Sun it has ___. When our place faces away it has ___. Day turns to night because Earth ___. **Staff I ☐ S ☐ C ☐**

Help: Check Reference A to see which side is lit and what movement carries the marker around.

3. Explain why darkness comes each evening without the Sun switching off.

Staff I ☐ S ☐ C ☐

Help: Think about what the globe did while the lamp stayed shining.

Optional participation-loop prompt: Share one explanation with an adult, hear one feedback step, and respond below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. In the North-Pole model, Place X is on the unlit right half of Earth. Is Place X in day or night, and what does Earth do so Place X reaches daylight?

2. How many hours does one full rotation of Earth take?

3. Why does the Sun not need to switch off for us to have night?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Supported

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt